WaiYan (Anson) Chan

Ph.D. Candidate / Graduate Research Assistant
University of Wisconsin-Madison
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Education

University of Wisconsin-Madison Expected Summer 2028
Ph.D. in Electrical and Computer Engineering
- Minor: Mechanical Engineering
- Thesis: Over-Actuated Axial Flux Permanent Magnet Motor Using PCB Stator
- Advisor: Lei Zhou

University of Wisconsin-Madison Sep. 2022 – Jun. 2024
M.S. in Electrical and Computer Engineering
- Thesis: Scaling of High-Speed Surface PM Bearingless Machines
- Advisor: Eric L. Severson

University of Wisconsin-Madison Sep. 2017 – Dec. 2021
B.S. in Mechanical Engineering
- Project: Compact Low Voltage Induction Machine with 3D Printed Housing
- **Dean's Honor List**

Research Experience

Precision Mechatronics and Control Lab Aug. 2024 – Present
Graduate Research Assistant
- Conduct design and control research on an over-actuated, lightweight axial-flux motor, leading the electromagnetic design and fabrication of a PCB-based stator and Halbach-array rotor for high power density rotorcraft/eVTOL applications.

Severson Research Group Aug. 2022 – Jun. 2024
Graduate Research Assistant
- Additively Manufactured (AM) Stator Housing for High-Speed Bearingless Generator
  - Funding Agency: *U.S. Department of Energy, DE-EE0009138*
  - Designed and fabricated AlSi10Mg AM motor housing with integrated cooling channels for a 100kW, 80kRPM twin bearingless generator for a combined heat and power (CHP) generation system.
- Bearingless Machines (BSPM) for Aerial E-Turbocharger Application
  - Funding Agency: *U.S. Army Research Laboratory, W911NF-20-2-0181*
  - Conducted structural and modal analysis to validate rotor design by identifying critical speed and rotor stress.
  - Modeled and conducted FE analysis of an 8kW 4-DOF twin-stator BSPM, along with component fabrication.
  - Facilitated the development of multi-physics modeling framework for BSPM and created Python scripts for evaluating machine constants and coil inductances in JMAG.

Wisc. Electric Machine & Power Electronic Consortium Sep. 2019 – Dec. 2021
Undergraduate Research Assistant
- Prototyped a desktop size dynamometer setup capable of characterizing electric machines rated up to 10 N-m of torque and 3000 RPM. Reduced the original setup volume by 35%.

- Collaborated with a team of graduate researchers to prototype a modular terminal box for powering a 200 kW NASA prototype machine. Responsible for sourcing parts and creating CAD models.
- Fabricated BSPM machines by producing stator slot windings and machining components used in the assembly.

Teaching Experience

University of Wisconsin-Madison — Jan. – May. 2026
Teaching Assistant, ME/ECE 577 Automatic Controls Lab

University of Wisconsin-Madison — Sep. – Dec. 2025
Teaching Assistant, ECE 342 Electronic Circuits II

Industry Experience

Motibera, Inc.
Motor Design Engineer Intern — May. - Aug. 2025
- Designed and analyzed a novel bearingless electric motor topology with axial force capability, leading electromagnetic simulation in JMAG and CAD modeling of the prototype, with a patent filing in progress.
- Performed electromagnetic modeling and control oriented analysis of bearingless motor systems, co-authored a conference paper and developed internal tools to support design optimization studies.

Milwaukee Tool
NPD Mechanical Design Engineer — Jan. - Aug. 2022
- Conducted FE analysis on BLDC machines of various sizes and winding configurations; recommended optimal designs for product development using a Pugh Matrix to support informed and cost-effective decision-making.
- Assessed performance requirements of competitor power tools by collecting and analyzing motor thermal characteristics and power output data.

Journal Publications

J1 T. Noguchi, **W. Chan**, N. Petersen, L. Rapp, and E. Severson, "Opportunities to enhance sCO_2 power cycle turbomachinery with bearingless motor generators," Solar Compass, vol. 12, Dec. 2024, Art. no. 100094 (*DoE Award DE-EE0009138.*)

Conference Publications

C6 **W. Chan** and L. Zhou, "Over-Actuated Axial Flux Permanent Magnet Motor using PCB Sator," 2026 IEEE Energy Conversion Conference Congress and Exposition (ECCE), Vancouver, BC, Canada, 2026 (Accepted).

C5 **W. Chan** and L. Zhou, "Designing for Simplicity: A Novel Segmented-Stator Axial Flux Permanent Magnet Motor with Tape-Wound Cut Cores," 2025 IEEE Energy Conversion Conference Congress and Exposition (ECCE), Philadelphia, PA, USA, 2025, pp. 1-7.

C4 N. Petersen, **W. Chan**, and A. Khamitov, "Thrust Control via Rotor Skewing in Integrated Twin Bearingless Motor," in 19th International Symposium on Magnetic Bearings (ISMB19), Otsu, Shiga, Japan, Aug. 22, 2025.

C3 **W. Chan**, T. Noguchi and E. L. Severson, "Scaling of High-Speed Surface PM Bearingless Machines," 2024 IEEE Energy Conversion Congress and Exposition (ECCE), Phoenix, AZ, USA, 2024, pp. 5631-5638 (*US ARL Cooperative Agreement W911NF-20-2-0181.*)

C2 T. Noguchi, N. Petersen, **W. Chan**, L. Rapp, E. Severson, "Bearingless Motor/Generator Applications in sCO_2 Power Cycles," The 8th Internationl Supercritical CO_2 Power Cycles Symposium, San Antonio, TX, USA, 2024 (*DoE Award DE-EE0009138.*)

C1 T. S. Slininger, **W. Chan**, E. L. Severson and B. Jawdat, "An Overview on Passive Magnetic Bearings," 2021 IEEE International Electric Machines & Drives Conference (IEMDC), Hartford, CT, USA, 2021, pp. 1-8 (*NSF Grant 1938610.*)

Invention Disclosures

I1 N. Petersen, **W. Chan**, and A. Khamitov, "Integrated Bearingless Motor with Axial and Radial Magnetic Levitation" Co-inventor, U.S. Provisional Patent, filed Aug. 15, 2025 (assigned to Motibera, Inc.)

Oral Presentations

O1 *Scaling of High-Speed Surface PM Bearingless Machines*, IEEE Energy Conversion Congress and Exposition (ECCE), Phoenix, AZ, USA, 2024.

Poster Presentations

P1 *Designing for Simplicity: A Novel Segmented Stator Axial Flux Permanent Magnet Motor with Tape Wound Cut Cores*, IEEE Energy Conversion Congress and Exposition (ECCE), Philadelphia, PA, USA, 2025.

Award and Honors

Best Presentation Award 2025
19th International Symposium on Magnetic Bearings (ISMB19), Otsu, Japan

Skills

Hands-on: Electric Machine Fabrication, Component Machining and Assembly, CAD Design and Drafting, GD&T
Tools: Python, Git, SolidWorks, NX, ANSYS, MATLAB / Simulink, Altium, LabView, MAGNET, FEMM, JMAG, LaTeX

Certifications

Fundamentals of GD&T ASME Y14.5, Applied Geometrics, Inc. Apr 2022